Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/08/2017
PI Name:	Wu, Honglu Ph.D.		
Project Title:	miRNA Expression Profiles in Cultured Human Fibroblasts in Space		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Comments:	PI email changed from honglu.wu-1@nasa.gov after PI retirement from NASA.		
Project Type:	Flight	**Solicitation / Funding Source:**	2011 Space Biology NNH11ZTT002N
Start Date:	10/01/2012	**End Date:**	03/31/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 3/31/2016 per A. Chu/ARC (Ed., 10/28/15) NOTE: End date changed to 9/30/2015 per A. Chu/ARC (Ed., 2/24/15) NOTE: End date changed to 3/31/2015 per A. Chu/ARC (Ed., 4/11/14)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mangala, Lingegowda (University of Houston) Zhang, Ye (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microgravity or an altered gravity environment from the static 1g has been shown to influence global gene expression patterns and protein levels in cultured cells or animals, but it is unclear how these changes in gene and protein expressions are related to each other or are related to other factors regulating such changes. Recent advancement in the field of molecular biology revealed that a different class of RNA, the small non-coding microRNA (miRNA), can have a broad effect on gene expression networks by mainly inhibiting the translational process. In this experiment conducted on the International Space Station (ISS), we propose to test the hypotheses that miRNA profiles will be altered in the space environment, and that cellular responses to DNA damage in space are different from those on the ground.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	On April 18, 2014, our Micro-7 payload was launched at Kennedy Space Center. The cells arrived on the International Space Station on April 22. Our payload consisted of two experiments. In the experiment aimed at studying RNA/miRNA expression changes in non-dividing cells in space, human fibroblast cells were fixed with RNAlater2 on April 25 and on May 5, which corresponded to 3 and 14 days after reaching orbit, respectively. The cells were also fixed with paraformaldehyde for immunohistochemistry staining. For investigations of altered cellular responses to DNA damage, human fibroblast cells were treated with bleomycin on April 22 (Day 3). After a 3-hour treatment, the cells were fixed with RNAlater2 for RNA/miRNA analysis and with paraformaldehyde for immunohistochemistry staining. Summary of Present Micro-7 Experimental Results RNA/miRNA experiment: Our Micro-7 payload consisted of two experiments. In the experiment aimed at studying RNA/miRNA expression changes in non-dividing cells in space, cells in 4 BioCells in space and on the ground, respectively, were fixed with RNALaterII on Day 3 and again on Day 14. Hierarchical clustering analysis indicated expressions of 1924 genes with a false discovery rate FDR<0.08 for both the flight and ground samples on Day 3. These genes were well clustered between the quadruplicates of the flight and ground samples. However, no significant differences in gene expressions between the ground and flown cells were found on Day 14. A PCA of 47k probes showed that each of the three components accounted for 19.5%, 13.5%, and 9.6%, respectively, of the data set variance. Our PCA mapped a total of 42.6% of the data set variance. Of the genes analyzed, 170 were found to have significant expression changes between space and ground conditions in the Day 3 samples. This compares to only 3 genes showing similarly significant expression changes between space and ground by Day 14. Additionally, none of the 170 genes identified in the Day 3 comparison overlapped with genes identified in the Day 14 comparison. Of these 170 genes, 117 have known protein coding functions. For identification of pathways and networks in the flown cells on Day 3, the list of 170 genes and 29 miRNAs with their respective expression changes were analyzed with the Ingenuity Pathway Analysis (IPA) tool. The top upstream regulators that were identified included Vegf, HGF, Ca2+, and RELA. In addition to the above four top regulators, MAPK8 and TNFRSF10B were among the nodes. The miRNAs that were identified in the network included the Let-7 family, miR-29b, and miR-708. This part of the project has been completed and a manuscript has been published in the June, 2016 issue of FASEB J. DNA damage response experiment For investigations of the cellular response to DNA damage from bleomycin treatment, human fibroblast cells were treated with bleomycin at a final concentration of 1 microg/ml on April 22 (Day 3). After a 3-hour treatment, the cells were fixed with RNAlater2 for RNA/miRNA analysis and with paraformaldehyde for immunostaining. In flight, there were 52 genes whose expressions were significantly altered as a consequence of bleomycin treatment during flight experiment. On the other hand, in ground cells treated with bleomycin, 42 genes showed significant expression changes. Among these genes, 22 were common in both contrasts. Analysis of pathways using the IPA software demonstrated that p53 was the primary pathway responding to the bleomycin treatment in both the flown and ground cells. In addition, TP73 and RELA were involved in both the flown and ground cells in response to bleomycin treatment. TP53, TP73, and RELA were among the top 5 upstream regulators identified with IPA. Including the indirect interactions, our data with bleomycin produced similar responses as other chemical toxins that induce DNA damages such as hydrogen peroxide, cisplatin, and mitomycin C.

Bibliography Type:	Description: (Last Updated: 07/15/2026)
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